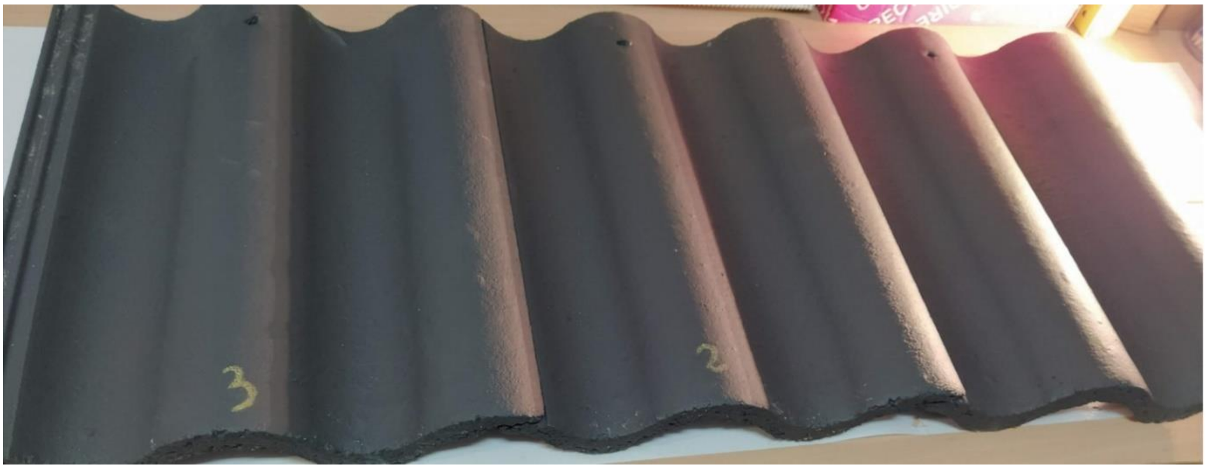
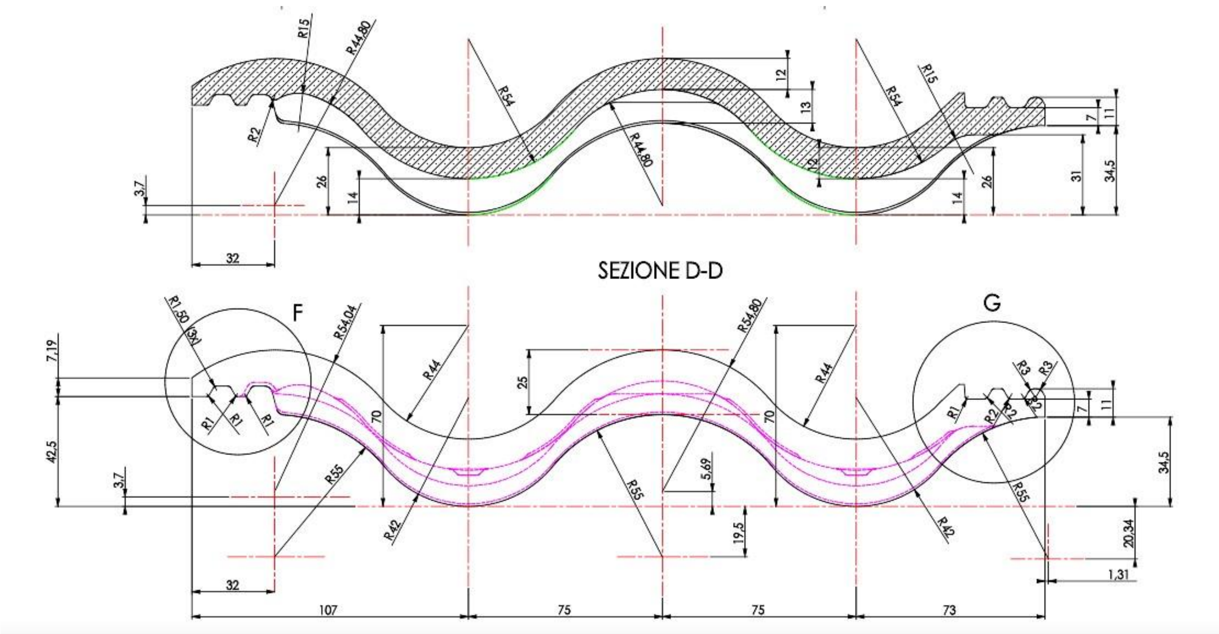
ACKERSTEIN 09/02/22

Due to the breakage of **SEA WAVE** tiles on the roofs, a video call was made on the 9th of February in which a request was made to ACKERSTEIN to check the thickness of the tiles and to send 12 tiles to Spain and Italy.

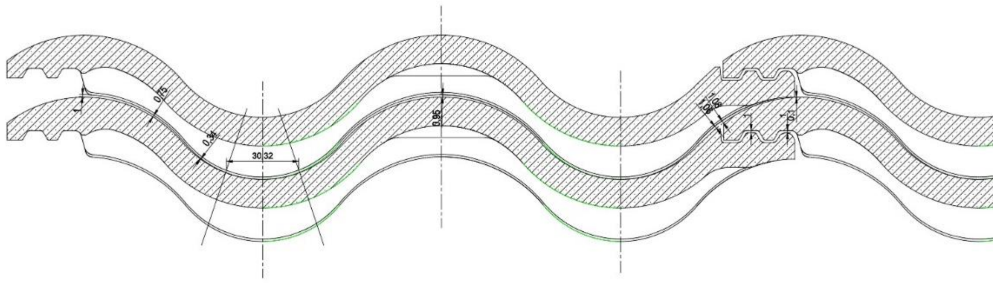
In Spain, I work with 3 tiles from each group for the checks.



The **SEA WAVE** tiles, according to the drawing, must have these measurements.



It is necessary to achieve



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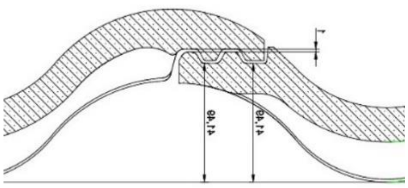
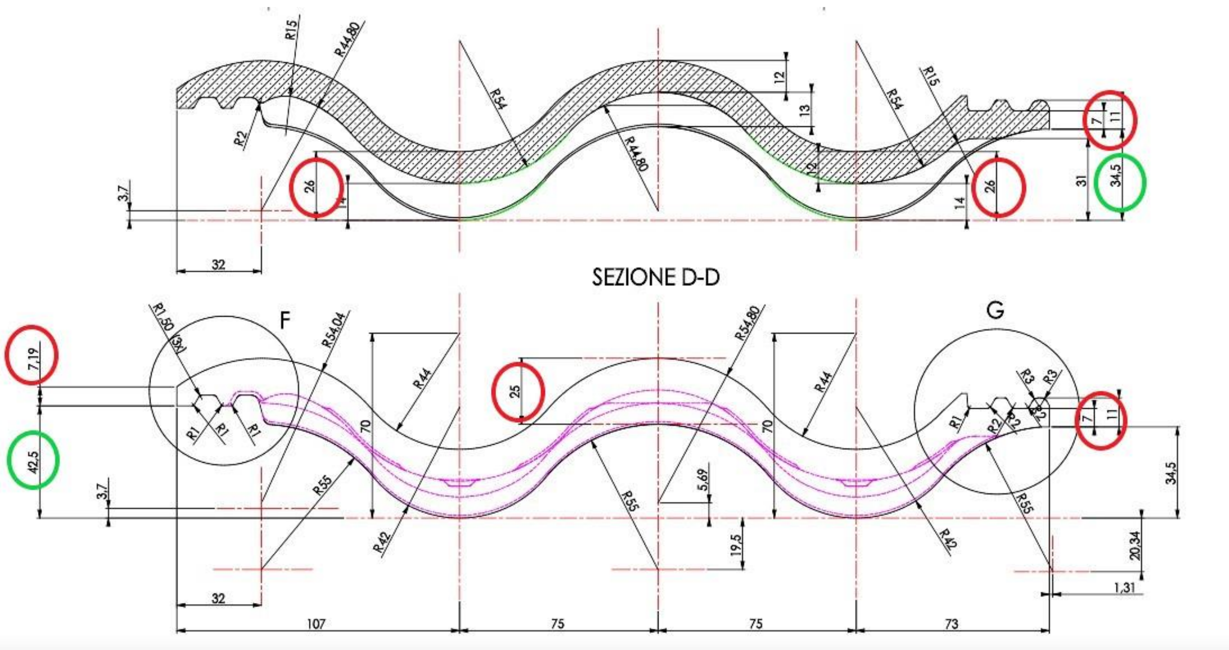
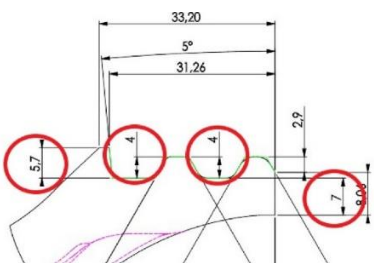
a good coupling between the tiles on the roof tile.



The most important measurements are in the "**RED** circle" given by the slippers and in the "**GREEN** circle", given by the aluminium mould/pallet:



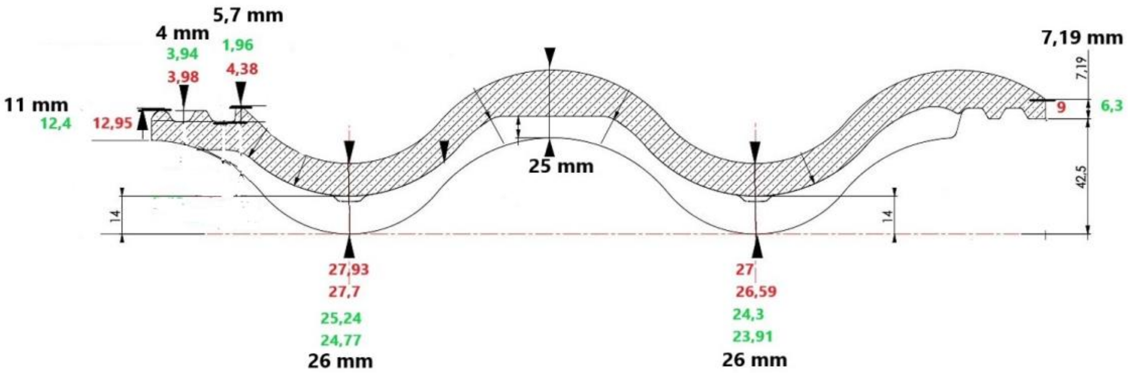


Here below the results of the checking of the 6 tiles sent to Spain.

Weights:

3.939 gr. 4.019 gr. 4.002 gr. 4.880 gr. 4,890 gr. 4.916 gr

Measurements on concrete roof tiles



On aggregates indicate the optimal granulometry in Spain and the specific objectives of the sand were.

The optimum sand should be of the following granulometry.

nº Sieve.		Size in mm.	%	% of the total
	4	4,76	0	0
	8	2,38	10	10
	12	1,7	17	7
	116	1,19	27	10
	30	0,59	57	30
	50	0,297	78	21
	100	0,149	94	16
	200	0,074	97	3
Powder		0,05		3
				100

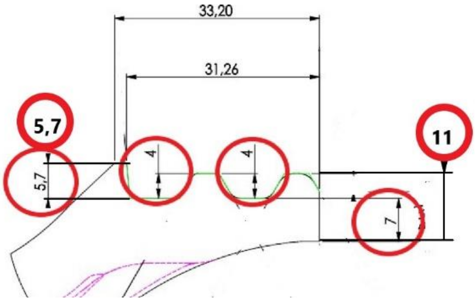
SAND PREFIXED OBJECTIVES:

Fineness Modul ....(2,7 a 3 %)
Fines 1 (5%)……………………..(2 a 5 %)
Sand Equivalent………… …..(80 al 85 %)

ACKERSTEIN 21/02/22

In the factory, thickness checks are made on the fresh tiles and depending on the thicknesses in the assembly.

I suggested to adjust the thickness of the assembly to 5.7 mm and 11 mm.



SETTING OF SLIPPER WITH THE SLIPPER AND GAP WITH SMALL ROLLER		
ACTIONS / NOTES	**PLAN**	**ACTIONS / NOTES**
		5) The standard setting of the height of the small slipper compared to the slipper is for SEA WAVE 5,7 mm
		6) With a venier verify the same height of the small slipper compared to the slipper in its front and rear edges, see image, to check the right inclination with the slipper.
	HEIGHT BETWEEN THE ROLLER AND THE SMALL ROLLER.	7) Another way to start the setting is to measure the height between the roller and the small roller. (See image Red circle). This can be the measure to start the setting of the small slipper height compared to the slipper. 8) In any case, verify if the profile of the extruded roof tile is correct or not, and operate consequently

Diameter Small Roller 119 - Diameter Big Roller 107,65 mm = 11,35 : 2 = **5,7 mm**

The granulometry of the aggregates indicated can be optimal in Spain with the specific objectives. In Israel there are very different sands that can give very good results but in some of the sands the moisture they bring is a problem.

MIXER NEW RECIPE

SAND 430 Kg.

BAZALT 170 Kg.

CEMENT 52,5 175 kg.

WATER 80 liters.

MOISTURE :

8,77 % (8,5 % / 9 % in SPAIN). Depends on the aggregates**.**

DRY TILE

4,576 kg……4,586 kg.

Objective…………4,398 kg. Depends on the aggregates.

Measures :

Center of tile………. 13,23 + 14 (27,23)…….12,75+ 13 (25,75)…… 13,4 + 14 (27,4)

"Objective"………..26 mm(-1,23)……..25 mm(-0,75)……….26(-1,4) mm

Interlock……………….11,18 (11)………………5,3 /6 (5,7))…………. 8,75 (7,9)

"Objective"………..11 mm(+0,18)…….5,7 mm. (-0,4)………7,9(-0,85) mm

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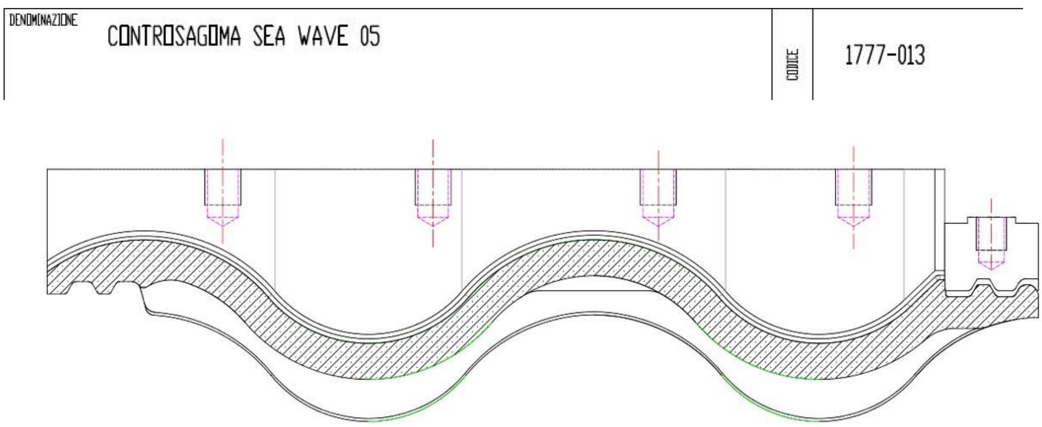
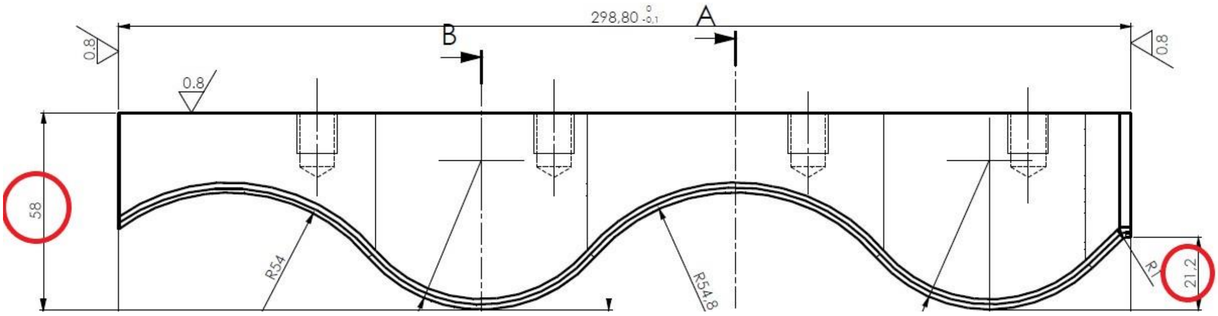
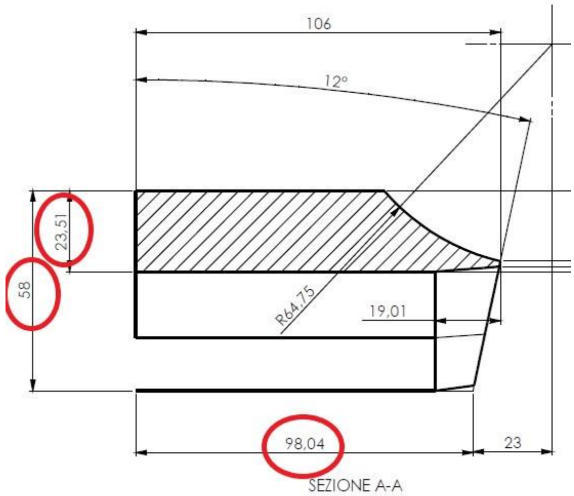
Regarding the problem of the possible difference in the profile of the steel slipper (May-October) and the widia slipper / tungsten (November-February), the drawings of the two slipper are requested.

The steel slipper corresponds to the drawing **1777-013.**



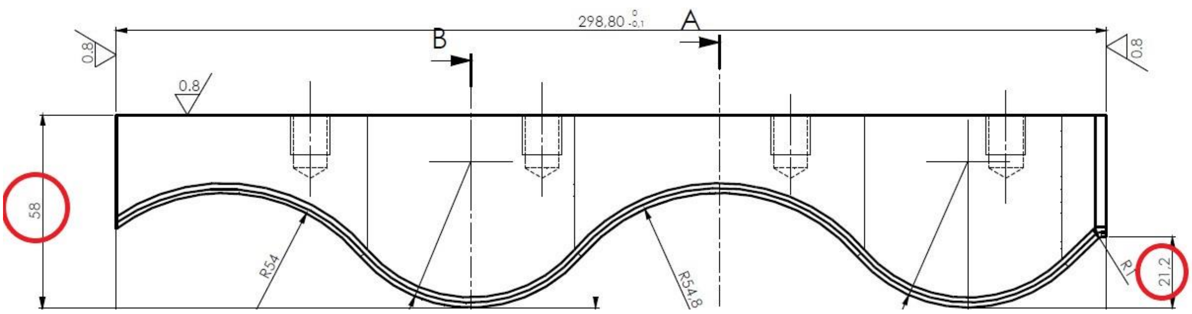


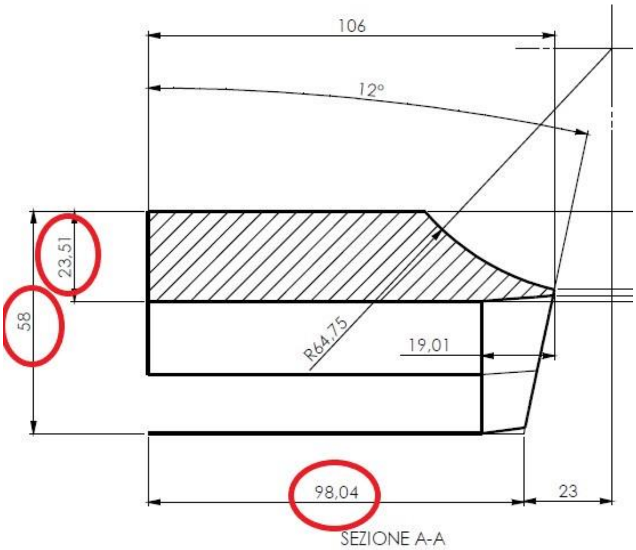


The widia/tungsten slipper corresponds to the drawing **1777-023.**

DENOMINAZIONE CONTROSAGOMA SEA WAVE 05 WIDIA	CODICE	1777-023

These are the same dimensions “as the steel slipper”.





Meeting with Adam to evaluate the various types of the available sands in order to further adjust the mix to the granulometric curve presented.

Tests with different sands are planned for 22 february 2022.

ACKERSTEIN 22/02/22

First test of new aggregates with more basalt sand, it gets a better surface appearance, but it comes wetter and the amount of water has to be reduced.

MIXER RECIPE

SAND 400 Kg.

BAZALT 200 Kg. (High Humidity)

CEMENT 52,5 175 kg.

WATER 80 liters. (75/80)

FRESH TILE

Fresh tile Measure:

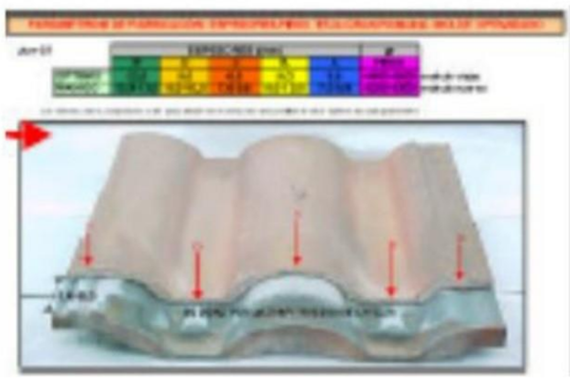
4,780 kg. (7480 – 2,700 mould)

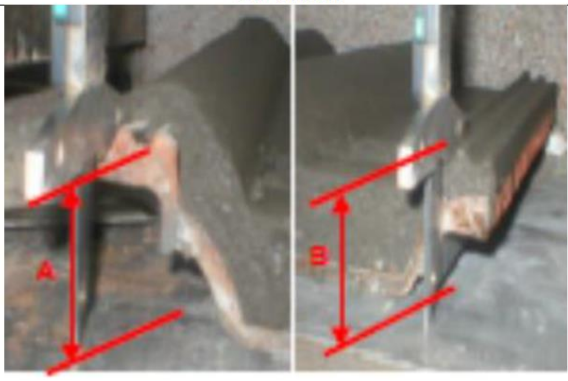
TEST WITH SEA SAND

Second test with sea sand, but being that was not possible to control the excess of humidity generated with the sands, the test has stopped immediately.

I suggested a daily check of the weights and thicknesses of the tiles, every 1 or 2 hours, that should be carried out with the new measurement tools.

Prepare the wet tile to measure its thickness according to the image.		Tools used: -Probe – Digital "caliper" -6 x 20 x 200 mm calibrated bar. -10 mm cylindrical bar 200 mm -Spatula
In order to measure points B, C, D and E we shall use a calibrated bar in order not to deform or vary its measurement. Once measured, we shall deduct the 6mm of the bar.		Take into account the perfect alignment of the caliper with the calibrated bar and its perpendicularity to the mould.
In order to measure "A" we shall use a cylindrical bar so as not to deform the concrete or vary its measurement. Once measured, we shall deduct the 10 mm		Take into account the perfect alignment of the caliper with the cylindrical bar and its perpendicularity to the mould.

Note the results in the extruder control sheet	
Adjust weights and thickness to what the central laboratory sets out. If there are any variations in the measurements, adjust the extruder head and repeat steps 1- 9 of this procedure.	

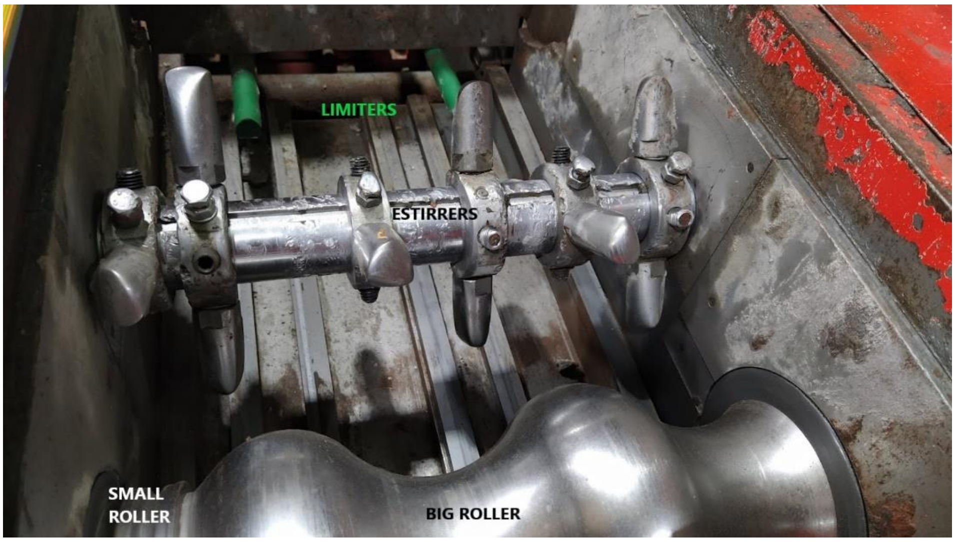
<table>
<tr><th>STEPS</th><th>DRAWING</th><th>KEY POINTS</th></tr>
<tr><td>Check that the “assemblies” meet the installation standards. Measure point “A” and check it is bigger than “B”.</td><td>
</td><td>This point is critical.
If B >= A,
the weight shall be immediately corrected and lower the extruder head. If we don’t get A>B, then change the extruder head.</td></tr>
</table>

I suggested in the extruder place one more stirrer in front of the junction of the big and small slipper.

After control of the results dry tile in the laboratory, the adjustment of the thickness of the tile about production 23 february, I suggested, to lower the thickness on both sides of the big slipper by <u>“ 0.5 mm”.</u>

<u>ACKERSTEIN 23/02/22</u>

In the morning, revision of the extruder box :



A new stirrer is placed in the interlock side, shifted 5 mm to the left, all stirrer positions are identified, the lengths of the limiters and the roller diameters are checked.

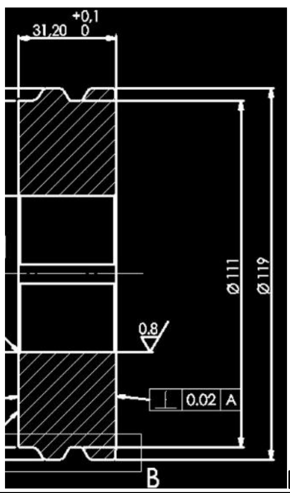
During extruder box set-up, I have identified that for better settings of rollers and slipper a metal bar should be used between the two guides of the bench.

I suggested the use of a metal bar (300 x 100 x 5 mm) to make all the adjustments the extruder needs.

Video UNO 10 Extruder settings has been provided to customer, with addiotional notes in a separate document.

UNO10 Extruder settings

The small roller has excessive wear on the large roller side.



D

I suggest a daily CONTROL of the extruder situation (ROLLERS,SLIPPERS, STIRRERS, LIMITERS) that is necessary to improve the tile forming process.

	PROCESS SHEET	Code Date: 090222 Review number: Page: 1

FACTORY: ACKERSTEIN
SECCION : EXTRUDER BOX
PROCESS : EXTRUDED / TILE CONFORMATION Nº 1

PROFILE	Nº	PARAMETERS	NOMINAL VALUE	MINIMUM VALUE	MAXIMUM VALUE	UNITS	NOTES
SEA WAVE - HEAD	1	Roller speed.				r.p.m.	ALL THESE VALUES
	2	Stirrer speed.				r.p.m.	ARE ESTABLISHED FOR
	3	Roller height.				mm	PRODUCTION SPEED OF
	4	Slipper entry height				mm	TILES PER MINUTE
	5	Slipper exit height.				mm	
	6	Limiters length . (steel / policord))				mm	
	7	New roller diameter.				mm	
	8	Used roller diameter.				mm	
	9	Height distance between big and small slippers				mm	
	10	Maximum roller temperature				ºC	
	11	Tiles per minute				t.p.m.	

It is necessary to identify the duration of the "Big and Small Roller steel" and "Big and Small Slipper widia", to avoid making the concrete tile incorrectly

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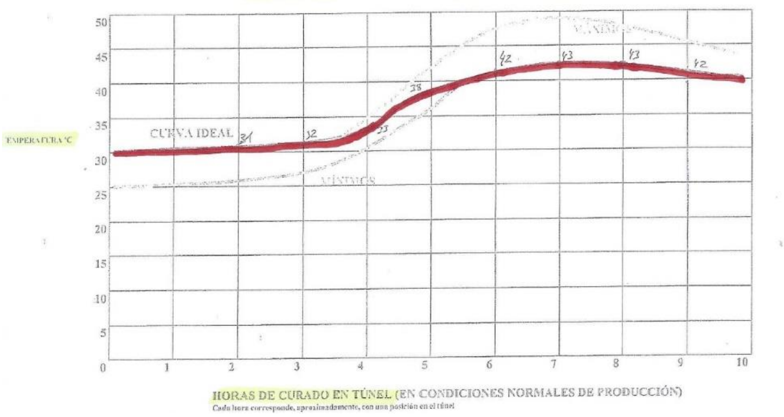
I suggest the CONTROL and monitoring of the mixer.

	PROCESS SHEET	Date: 090222 Review number: Page: 1

FACTORY: ACKERSTEIN
SECCION : MIXER
PROCESS : MIXTURE

PROFILE	Number	PARAMETERS	NOMINAL VALUE	MINIMUM VALUE	MAXIMUM VALUE	UNITS	NOTES
		DOSAGE					
ALL PROFILES FINISHES AND COLORS	1	Sand fine Storage Hopper 1				kg	
	2	Sand groos Storage Hopper 2				kg	
	3	Sand compensation moisture				kg	
	4	Moisture sand.				%	
	5	CEM II / A-V 52,5 Portland				kg	
	7	Pigment				kg	
	8	Waterproofing				kg	
	9	Water				Litres.	
	10	Total mixture weight				kg	
	11	Sand humidity control time				seconds	
	12	Sand / Pigment mixture time.				seconds	
	13	Moisture mixture.				%	
	14	Dry mixture time.				seconds	
	15	Wet mixture time				seconds	
	16	Total mixture time.				seconds	
	17	Theorical mixture time.				seconds	
	18	Theorical cicle time.				seconds	

Meeting with Charlie and Adam about the curing chambers, the cycle that is used in Spain and the possibilities for changes that can be made to the current curing chambers.



For the Chamber Curing Process, **I suggest :**

a) **Placing a system of pipes and extractors on the roof of the chamber to extract excess humidity above 98 %.**
b) **Install hot air pipes with diffusers in the lower parts of the chambers to cause an envelope in all the tile rackers, so that they try to equalise the chamber temperature at all heights, thereby ensuring better curing.**

For the Paint Application Process, **I suggest** :

The heating of the paint to better polymerise the application, a drying tunnel at the outlet of the application and a line of high-flow downstream fans to better distribute the paint in the central parts of the concrete tile.

ACKERSTEIN 24/02/22

Fresh Tiles Production 240222

Measures :

Center of tile.......... 13,15 + 14 (27,15).......12,75+ 13 (25,75)...... 13,4 + 14 (27,4)

"Objective"...........26 mm(-1,15).........25 mm(-0,75)..........26(-1,4) mm

Interlock...................11,5 (11)......................6 (5,7))................... 8,75 (7,9)

"Objective"...........11 mm(+0,5).......5,7 mm. (-0,3).........7,9(-0,85) mm

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Measuring tiles in Laboratory -DRY TILE Production 230222

(17 hours after Curing) Strengh : **2.415 N.**

Weight: 4,628 kg.

Measures :

Center of tile.......... 13,2 + 14 (27,2).......12,75+ 13 (25,75)...... 13,7 + 14 (27,4)

"Objective"...........26 mm(-1,2).........25 mm(-0,75)..........26(-1,7) mm

After check of the results on dry tiles, I suggested to lower the thickness only on the "wave" side of the big slipper by " 0.5 mm".

Interlock.....................11 (11)......................6 (5,7))..................... 8 (7,9)

"Objective"...........11 mm()...............5,7 mm. (+0,3).........7,9(-0,1) mm

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At the moment the measurements of the interlock are correct according to the drawing, regarding the measurements in the centre of the tile, I suggest that they should be reduced from 13 mm to 12.5 mm and maintain a minimum of 12.5 mm in the centre of the tile, as all

the sections are not equal in their compactions, so a minimum of "12.5" mm of section should be admitted.

Once the 13 mm / 12.5 mm / 13 mm measurements are stabilised, after 4 weeks of production and it can be demonstrated by the process sheets extruder and tests, that their weights and resistances are maintained, the following test can be considered to reduce the size of the tile to 12.5 mm / 12 mm / 12.5 mm.

All changes of extruder settings should be made very slowly to check the results as sands and moistures change the tile dimensions and properties.

Meeting 14:00

An assessment is made of what has been done and what has been commented on during the period of action from 21 to 24 February 2022 in the factory and possible future actions.

a) The details of the revisions of the roof tiles sent to Spain were assessed, regarding the thickness and appearance of the roof tile surfaces.

There is a requirement to fit the tiles on the roof, which, if the measurements of the assembly are not adequate, can cause breakage of the tiles on the roof due to poor support of one tile with the other. There is a big difference between the reference thicknesses according to the drawing and the actual thicknesses of the tiles.

The oscillations in the weights and thicknesses are caused by the sands, as they change the density of the concrete, but the weights with the same mixture of sands should not vary more than 50/100 grams.

Regarding the surfaces of the tiles, in my assessment I have had missing information on when they could be generated, in the extrusion or by paint applications.

b) It has been commented on the profile of the tile produced from (May to October) and that of (November to February) and it has been stated that on the drawings provided 1777-013 and 1777-023, that there is a constant profile SEA WAVE in the big and small steel and widia slippers supplied.

d) The values of the optimum sand in Spain and the possible mixtures in the sands of Israel have been assessed, therefore some tests have been planned and carried out to identify if there is a possible better mixture of the current sands.
The need for a good screening of the sand has been identified by finding in the body of the tile at the exit of the extrusion large stones and conglomerates of the dune sand, which generate expansions on the surface of the tile at the exit of the big slipper.

In my experience with screens, I have had the best results with LIWELL screens and with mixer with EIRICH RV.

The following components must be available for sand installations: a moisture controller in the lower parts of the silos to deliver the kilos of sand we need, discounting the kilos of water for the moisture of the sand, and a moisture controller in the mixer.

e) Regarding the wear of the extruder elements, I have indicated an estimated minimum and maximum duration of wear, which is always determined by the types of sand used.

The table on the estimated wear of each part of the extruder has been provided to Adam.

A possible change to be considered would be the use of the small widia roller with the aim of changing the shaft as little as possible with the new rollers.

On average, a complete change of roller assembly for maintenance and for replace the parts with new ones takes around 3-4 hours, you can consider an entire morning to be on the safe side.

Further information on others customers that produce SeaWave tiles' criterias for replacing slippers and rollers will be given once collected.

f) In my opinion, in order to make progress in the production of a QUALITY tile, it is necessary to carry out rigorous PROCESS CONTROLS of the tile from the mixer, extruder, oil application, paint application, racker stacking, tile curing in the chambers, racker unstacking, demoulding, dry paint application, paint drying, stacking, packaging and palletising, etc.

The more processes can be controlled, the more process improvements can be planned, which will lead to improvements for a higher QUALITY roof tile and higher customer satisfaction.

g) About the extruder the current operating conditions must be known, I have only identified the speed of the roller 160 r.pm and that the possible height of the big slipper is 32 mm, there is no further precision knowing the inclination of the slippers.

The big roller already has a wear in its diameters of 2 mm and the small roller already has a deformation in the plane between the two rollers.

h)) It is insisted on the necessity to control the critical adjustments in the measures in the assembly 5,7 / 11 mm and in the parts of the centre of the tile (13 / 12,5 / 13 mm) and lateral wave of 7,9 mm to assure the best support of the tiles in the roof.

Considerations on the future of the thicknesses in the tile, it is necessary to maintain the measures of the possible assembly of the tile and to progress in reducing the measures of the thickness of the centre of the tile, from my point of view it can be possible to arrive at the values of 12,5 / 12/ 12,5 mm (according to mixture of sands) can arrive at its limit, assuring that these thicknesses maintain the good values of resistance to flexion and support the effort on the transit of people on the roof.

i) We reviewed approaches to improvements in the curing of the chambers and assisting in the drying of the paint on the dry roof tiles.

j) Finally, assemblies were made with tiles produced last year and this week's tiles, it was found that the old tiles do not support enough for the measurements of the assembly as the ones made this week.

In my opinion, tiles from different productions should not be mixed and the longer the time they spend curing in the yard, the more the possibility of breakage in the roof will be reduced.

ASHDOD 24/02/22